Handout for part 2 of Transport Problems

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This handout is intended to accompany the mini-lecture covering Sections 10.1.6 to 10.1.8 in Larson/Bengzon.

It collects the main results and gives more detailed proofs (particularly for existence/uniqueness via Lax–Milgram and the a priori error estimate), and summarizes the implementation ideas for the numerical example.

1 Model Problem and GLS Formulation

Lets start by recalling the stationary transport equation (Larson/Bengzon 10.1):

$$\begin{aligned} -\epsilon\Delta u + b\cdot\nabla u &= f, \quad \text{in } \Omega \\ u &= 0, \quad \text{on } \partial\Omega \end{aligned} \tag{1}$$

Where:

- $\varepsilon > 0$ is a small parameter denoting the diffusion coefficient
- b is a given vector field with $\nabla \cdot b = 0$
- f is a given function

Operator form and standard weak formulation

If we define the differential operator as follows:

$$L := -\epsilon\Delta + b\cdot\nabla$$

we can rewrite (1) as:

$$Lu = f \tag{2}$$

Now let $V = H_0^1(\Omega)$ and equip it with the usual H^1-norm $\|\cdot\|_{H^1}$

Multiplying (1) with $v \in V$ and integrating by parts yields the standard bilinear and linear forms:

$$\begin{aligned} a(u,v) &= \epsilon(\nabla u, \nabla v) + (b\cdot\nabla u, v) \\ l(v) &= (f, v) \end{aligned} \tag{3}$$

Hence, the weak formulation from part 1 becomes:

Find $u \in V$ such that

$$a(u,v) = \ell(v) \quad \forall v \in V.$$

The standard Galerkin finite element method then restricts u and v to a conforming finite-dimensional subspace $V_h \subset V$.

Modifications for Galerkin Least Squares (GLS)

The main idea of the GLS method is to add diffusion along the streamlines. The goal is to combine the stability from the least squares method with the accuracy of the standard Galerkin method. Thus we want to apply the test function $v + \delta Lv$ (with $\delta > 0$ being a suitably chosen parameter for maximal accuracy) to (2).

But before we can do this we have a problem. The second order derivatives of the test function v are unbounded across element boundaries and thus the term $-\epsilon\Delta v$ does not lie in $L^2(\Omega)$

To solve this, we define $\mathcal{K}$ as the mesh with elements $K \in \mathcal{K}$. Here the term $-\epsilon\Delta v$ is found in $L^2(K)$

On the discrete level, this leads to:

Find $u_h \in V_h$ such that

$$a_h(u_h, v) = \ell_h(v) \quad \forall v \in V_h, \tag{4}$$

where

$$a_h(u, v) := a(u, v) + \delta \sum_{K\in\mathcal{K}} (Lu, Lv)_{L^2(K)}, \tag{5}$$

$$\ell_h(v) := \ell(v) + \delta \sum_{K\in\mathcal{K}} (f, Lv)_{L^2(K)}. \tag{6}$$

We call (Lu, Lv) and (f, Lv) the GLS stabilization terms.

Since we only apply GLS stabilization to the interior of the elements. For continuous piecewise linear finite elements we have:

$$\Delta v = 0$$

Thus, on V_h the GLS formulation (5)(6) simplifies to:

$$a_h(u, v) = \epsilon(\nabla u, \nabla v) + (b \cdot \nabla u, v) + \delta(b \cdot \nabla u, b \cdot \nabla v) \tag{7}$$

$$\ell_h(v) = (f, v) + \delta(f, b \cdot \nabla v) \tag{8}$$

Here we see that this form looks alot like the standard bilinear forms from (3), but with the additional term

$$\delta\,(b \cdot \nabla u, b \cdot \nabla v)$$

that acts like added diffusion in the direction of the flow b and is often called a streamline diffusion term.

2 Existence and uniqueness via Lax-Milgram

We now want to prove existance and uniqueness of the solution u_h to (4)-(8) by showing coercivity and continuity of a_h, and continuity of ℓ_h before applying the Lax-Milgram theorem.

But to make our life a little easier we first define a new norm and a give a nice lemma.

2.1 GLS (energy) norm

We would like a norm that is more natural to use than $\|\cdot\|_{H_0^1(\Omega)}$, hence we define the following norm that is closely related to the bilinear form $a_h(\cdot,\cdot)$:

Definition 2.1 (GLS norm)**.**

$$|||v|||^2 := \varepsilon \|\nabla v\|^2_{L^2(\Omega)} + \delta \|b \cdot \nabla v\|^2_{L^2(\Omega)}$$

One thing to note is that if $\varepsilon = 1$ and $\delta = 0$ this is actually the energy norm that we have seen before in this course.

We will now verify that this more general version is indeed also a norm:

Proof.
It is straightforward to check that this is indeed a norm on $H_0^1(\Omega)$. We must show positivity, homogeneity and the triangle inequality.

- positivity: $|||v|||^2 = 0$ implies $\nabla v = 0$ and $b \cdot \nabla v = 0$, hence v is constant and zero on $\partial\Omega$, so $v = 0$;
- homogeneity and triangle inequality follow from those of the L^2-norms.

□

From the definition of the norm above we also get the following trivial inequalitys that we will use later.

Lemma 2.2 (Trivial inequalitys from GLS norm)**.**

$$\sqrt{\epsilon}\|\nabla v\| \leq |||v||| \tag{9}$$
$$\sqrt{\delta}\|b \cdot \nabla v\| \leq |||v||| \tag{10}$$

From now on, we shall view V_h as a finite-dimensional subspace of $(V, |||\cdot|||)$.

2.2 A quick lemma

We will use the following lemma in proving coercivity of a_h

Lemma 2.3 (Quick lemma)**.**
Suppose $b \in W^{1,\infty}(\Omega)^d$ *with* $\nabla \cdot b = 0$ *and* $v \in H_0^1(\Omega)$*. Then*

$$(b \cdot \nabla v, v) = 0.$$

Proof.
From the Gauss divergence theorem we have:

$$\int_{\partial\Omega} F \cdot n ds = \int_{\Omega} \nabla \cdot F dx$$

If we set $F = bv^2$ and plug in we get:

$$\underbrace{\int_{\partial\Omega} (b \cdot n) v^2 ds}_{\text{Term A}} = \underbrace{\int_{\Omega} \nabla \cdot \left(bv^2\right) dx}_{\text{Term B}}$$

- For **term A**: Since $v \in H_0^1(\Omega)$ we have that $v^2 = 0$ on $\partial\Omega$ and hence, we have that the entire term is equal 0. Hence, we get:

$$\int_{\partial\Omega} (b \cdot n) v^2 ds = 0$$

- For **term B**: We can rewrite it as:

$$\int_{\Omega} \nabla \cdot \left(bv^2\right) dx = \left(\nabla \cdot \left(bv^2\right), 1\right)_{L^2(\Omega)}$$

  Next, lets expand $\nabla \cdot \left(bv^2\right)$ using the product rule for divergence:

$$\nabla \cdot \left(bv^2\right) = (\nabla \cdot b)v^2 + b \cdot \nabla \left(v^2\right) = (\nabla \cdot b)v^2 + b \cdot (2v\nabla v) = (\nabla \cdot b)v^2 + 2vb \cdot \nabla v$$

  Inserting this into the inner product with 1 and using linearity of the inner product we get:

$$\left((\nabla \cdot b)v^2 + 2vb \cdot \nabla v, 1\right) = \underbrace{\left((\nabla \cdot b)v^2, 1\right)}_{\text{Term 1}} + \underbrace{(2vb \cdot \nabla v, 1)}_{\text{Term 2}}$$

  - For term 1 we have:

$$\left((\nabla \cdot b)v^2, 1\right) = \int_{\Omega} (\nabla \cdot b)v^2 dx = \int_{\Omega} (\nabla \cdot b)vv dx = ((\nabla \cdot b)v, v)$$

  - For term 2 we have:

$$(2vb \cdot \nabla v, 1) = \int_{\Omega} 2vb \cdot \nabla v dx = 2\int_{\Omega} (b \cdot \nabla v)v dx = 2(b \cdot \nabla v, v)$$

Putting it back together, with term A we end up with:

$$0 = ((\nabla \cdot b)v, v) + 2(b \cdot \nabla v, v)$$

Using the fact that $\nabla \cdot b = 0$ in Ω we have:

$$0 = 2(b \cdot \nabla v, v) \implies (b \cdot \nabla v, v) = 0$$

□

2.3 Coercivity of a_h

Coercivity follows quickly from the above lemma.

$$\begin{aligned}
a_h(v,v) &= \epsilon(\nabla v, \nabla v) + (b\cdot\nabla v, v) + \delta(b\cdot\nabla v, b\cdot\nabla v) \\
&= \varepsilon \|\nabla v\|^2 + (b\cdot\nabla v, v) + \delta \|b\cdot\nabla v\|^2 \\
&= \varepsilon \|\nabla v\|^2 + \delta \|b\cdot\nabla v\|^2 && \textit{(Lemma (2.3))} \\
&= 1\cdot |||v|||^2 && \textit{(Rewriting to GLS norm)}
\end{aligned}$$

2.4 Continuity of a_h

We start by splitting up the bilinear form into three parts as follows:

$$a_h(u,v) = A + B + C$$

where

- $A = \varepsilon(\nabla u, \nabla v)$
- $B = (b\cdot\nabla u, v)$
- $C = \delta(b\cdot\nabla u, b\cdot\nabla v)$

We will now show that each part is bounded and thus continuous.

2.4.1 For part A and C

We start with showing part A:

$$\begin{aligned}
A &\le \varepsilon \|\nabla u\| \|\nabla v\| && \textit{(Cauchy–Schwarz ineq.)} \\
&= \sqrt{\varepsilon}\|\nabla u\| \sqrt{\varepsilon}\|\nabla v\| && \textit{(split up } \varepsilon\textit{)} \\
&\le 1 |||u||| |||v||| && \textit{(GLS ineq. (9))}
\end{aligned}$$

And similarly for part C we have:

$$\begin{aligned}
A &\le \delta \|b\cdot\nabla u\| \|b\cdot\nabla v\| && \textit{(Cauchy–Schwarz ineq.)} \\
&= \sqrt{\delta}\|b\cdot\nabla u\| \sqrt{\delta}\|b\cdot\nabla v\| && \textit{(split up } \varepsilon\textit{)} \\
&\le 1 |||u||| |||v||| && \textit{(GLS ineq. (10))}
\end{aligned}$$

2.4.2 For part B

$$\begin{aligned}
B = (b\cdot\nabla u, v) &= -(b\cdot\nabla v, u) && \textit{(Integration by parts)} \\
&\le \|b\cdot\nabla v\| \cdot \|u\| && \textit{(Cauchy–Schwarz ineq.)} \\
&\le \|b\cdot\nabla v\| \cdot C_p \|\nabla u\| && \textit{(Poincaré Inequality)} \\
&\le \frac{1}{\sqrt{\delta}} |||v||| \cdot \frac{C_p}{\sqrt{\varepsilon}} |||u||| && \textit{(GLS ineq. (9) (10))} \\
&= \frac{C_p}{\sqrt{\varepsilon\delta}} |||v||| |||u|||
\end{aligned}$$

2.4.3 Putting it all back together

$$a_h(u,v) \leq (1+1+\frac{C_p}{\sqrt{\varepsilon\delta}})\,|||v|||\,|||u|||$$

Hence, we have continuity with

$$C_a = 2 + \frac{C_p}{\sqrt{\varepsilon\delta}}$$

2.5 Continuity of ℓ_h

Like with a_h we split up ℓ_h as follows:

$$\ell_h = A + B$$

with

- $A = (f, v)$
- $B = \delta(f, b \cdot \nabla v)$

2.5.1 For part A

$$\begin{aligned} A &\leq \|f\|\,\|v\| && \textit{(Cauchy–Schwarz ineq.)} \\ &\leq C_p\,\|f\|\,\|\nabla v\| && \textit{(Poincaré Inequality)} \\ &\leq \frac{C_p}{\sqrt{\varepsilon}}\,\|f\|\,|||v||| && \textit{(GLS ineq. (9))} \end{aligned}$$

2.5.2 For part B

$$\begin{aligned} B &\leq \delta\,\|f\|\,\|b \cdot \nabla v\| && \textit{(Cauchy–Schwarz ineq.)} \\ &\leq \delta\,\|f\|\,\frac{1}{\sqrt{\delta}}\,|||v||| && \textit{(GLS ineq. (10))} \\ &= \sqrt{\delta}\,\|f\|\,|||v||| \end{aligned}$$

Hence we have continuity of ℓ_h, and by the Lax-Milgram lemma there exist a unique solution to (4)-(8).

3 A Priori Error Estimate

We will now find the a priori error estimate for the GLS method. This will help us to determine the convergence (As $h \to 0$, the error $u - u_h \to 0$). We will also see that this method has a good rate of convergence, and in fact if we half the mesh size h, the error decreases (ideally) by a factor of $2^{3/2} \approx 2.83$.

As a reminder, here we have the exact solution u, the finite element solution u_h, the space V_h of continuous piecewise linears.

3.1 The GLS method is consistent

The GLS method (4) is consistent in the sense that it is satisfied by the exact solution u, that is:

$$a_h(u,v) = l_h(v), \quad \forall v \in V_h \tag{11}$$

In other words, if you plug the exact solution into the GLS stabilization terms (Lu, Lv) and (f, Lv), they cancel out.

3.2 Galerkin orthogonality

Now we subtract (11) from (4):

$$\begin{aligned} a_h(u, v) - a_h(u_h, v) &= l_h(v) - l_h(v) = 0 \\ \implies a_h(u - u_h, v) &= 0 \quad \forall v \in V_h. \end{aligned}$$

Thus we have obtained Galerkin orthogonality. Next comes the difficult part of the error analysis. Since $a(\cdot,\cdot)$ and $a_h(\cdot,\cdot)$ are not the same we do not have coercivity of $a(\cdot,\cdot)$ in the GLS norm as we have with $a_h(\cdot,\cdot)$. Therefor this error analysis is slightly different from the other error analysis we have done in this course.

3.3 Splitting the error with an interpolant

Let $\pi u \in V_h$ be the interpolant of u. We can now define the error as:

$$e := u - u_h = (u - \pi u) + (\pi u - u_h)$$

Now we apply the GLS norm and the Triangle inequality and get the following two terms that we will try to estimate in slightly different ways:

$$|||e||| \leq \underbrace{|||u - \pi u|||}_{\substack{\text{Term A} \\ \text{Estimate by} \\ \text{interpolation estimates}}} + \underbrace{|||\pi u - u_h|||}_{\substack{\text{Term B} \\ \text{Estimate by coercivity} \\ \text{and Galerkin orthogonality}}}$$

3.3.1 Estimating term B

First let $w := u_h - \pi u \in V_h$. From coercivity of a_h wrt. the GLS norm we have:

$$a_h(w, w) = |||w|||^2$$

From Galerkin orthogonality we have that $a_h(u - u_h, w) = 0$. Thus we can add this to $a_h(w, w)$ without changing the equality. Then using linearity of a_h we get:

$$a_h(w, w) = a_h(w, w) + a_h(u - u_h, w) = a_h(u - \pi u, w)$$

So now we have:

$$|||w|||^2 = a_h(u - \pi u, w)$$

We can now expand this term using (7) to get:

$$a_h(u - \pi u, w) = \underbrace{\varepsilon(\nabla(u - \pi u), \nabla w)}_{\text{Term I}} + \underbrace{(b \cdot \nabla(u - \pi u), w)}_{\text{Term II}} + \underbrace{\delta(b \cdot \nabla(u - \pi u), b \cdot \nabla w)}_{\text{Term III}}.$$

Let us estimate each of the three terms:

- For **Term I**:

$$\begin{aligned} I &= \varepsilon(\nabla(u - \pi u), \nabla w) \\ &\leq \sqrt{\varepsilon}\|\nabla(u - \pi u)\|\sqrt{\varepsilon}\|\nabla w\| && \textit{(Cauchy–Schwarz ineq.)} \\ &\leq \sqrt{\varepsilon}\|\nabla(u - \pi u)\|\,|||w||| && \textit{(GLS ineq. (9))} \\ &\leq C\sqrt{\varepsilon}h\|u\|_{H^2(\Omega)}\,|||w||| && \textit{(Interpolation estimate)} \end{aligned}$$

  With the interpolation estimate $\|\nabla(u - \pi u)\| \leq Ch\|u\|_{H^2}$

- For **Term II**:

$$\begin{aligned}
II &= (b\cdot\nabla(u-\pi u), w) \\
&= (n(u-\pi u), bw)_{L^2(\partial\Omega)} - (u-\pi u, \nabla\cdot(bw)) && \textit{(Integration by parts)} \\
&= -(u-\pi u, \nabla\cdot(bw)) && \textit{($u=\pi u=0$ on $\partial\Omega$.)} \\
&\le \|u-\pi u\|_{L^2(\Omega)}\|\nabla\cdot(bw)\|_{L^2(\Omega)} && \textit{(Cauchy–Schwarz ineq.)} \\
&\le Ch^2\|u\|_{H^2(\Omega)}\|\nabla\cdot(bw)\|_{L^2(\Omega)} && \textit{(Interpolation estimate)} \\
&\le Ch^2\|u\|_{H^2(\Omega)}\frac{|||w|||}{\sqrt{\delta}} && \textit{(GLS ineq. (10) and $\nabla\cdot b=0$)}
\end{aligned}$$

- For **Term III**:

$$\begin{aligned}
III &= \delta(b\cdot\nabla(u-\pi u), b\cdot\nabla w) \\
&\le \delta\|b\cdot\nabla(u-\pi u)\|\|b\cdot\nabla w\| && \textit{(Cauchy–Schwarz ineq.)} \\
&\le \sqrt{\delta}\,\|b\cdot\nabla(u-\pi u)\|\,|||w||| && \textit{(GLS ineq. (10))} \\
&\le \sqrt{\delta}\|b\|_{L^\infty(\Omega)}\|\nabla(u-\pi u)\|\,|||w||| \\
&\le C\sqrt{\delta}h\|u\|_{H^2(\Omega)}\,|||w||| && \textit{(Interpolation estimate)}
\end{aligned}$$

- Putting it all **back together**:

  If we look at the case of high convection $\varepsilon \le Ch$, then we have, $\delta = Ch$. Thus, the right hand sides of the three parts above, are all of order $h^{3/2}$. So in this regime, each term is $O\left(h^{3/2}\right)$. Summing them gives

$$|||w|||^2 = a_h(u-\pi u, w) \le Ch^{3/2}\,\|u\|_{H^2}\,|||w|||$$

  Divide by $|||w|||$ (assuming $w \neq 0$) to get

$$|||\pi u - u_h||| = |||w||| \le Ch^{3/2}\,\|u\|_{H^2(\Omega)}\,.$$

3.3.2 Estimating term A

We will now repeat the above estimate for $u-\pi u$ instead of $u_h - \pi u$, this time by using interpolation estimates.

$$e_I := u - \pi u \in H_0^1(\Omega)$$

Using the definition (7) of $a_h(u,v)$ we get:

$$a_h(e_I, e_I) = \underbrace{\varepsilon(\nabla e_I, \nabla e_I)}_{\text{Term I}} + \underbrace{(b\cdot\nabla e_I, e_I)}_{\text{Term II}} + \underbrace{\delta(b\cdot\nabla e_I, b\cdot\nabla e_I)}_{\text{Term III}}.$$

- For **Term I**:

  For the piecewise linear nodal interpolant πu, the standard interpolation estimates (assuming $u \in H^2(\Omega)$) are

$$\|\nabla e_I\| = \|\nabla(u-\pi u)\|_{L^2(\Omega)} \le Ch\|u\|_{H^2(\Omega)}$$

  Hence we get:

$$\mathrm{I} = \varepsilon\,\|\nabla e_I\|^2 \le \varepsilon C^2h^2\|u\|_{H^2}^2$$

- For **Term II**:

  Since $e_I \in H_0^1(\Omega)$ and $\nabla \cdot b = 0$, we can use lemma (2.3) and get:

$$\text{II} = (b \cdot \nabla e_I, e_I) = 0$$

- For **Term III**:

  For the streamline derivative, use boundedness of b :

$$\|b \cdot \nabla e_I\| \leq \|b\|_{L^\infty(\Omega)} \|\nabla e_I\| \leq C\|b\|_{L^\infty(\Omega)} h \|u\|_{H^2(\Omega)}$$

  Hence we get:

$$\text{III} = \delta \|b \cdot \nabla e_I\|^2 \leq \delta C^2 \|b\|_{L^\infty}^2 h^2 \|u\|_{H^2}^2$$

- Putting it all **back together**:

  Putting all the terms back together we have:

$$|||e_I|||^2 \leq Ch^2 \left(\varepsilon + \delta \|b\|_{L^\infty}^2\right) \|u\|_{H^2}^2$$

  Again we look at the case of high convection so $\epsilon \leq Ch$ and $\delta = Ch$. Thus we get:

$$|||e_I|||^2 \leq Ch^2 \cdot h \|u\|_{H^2}^2 = Ch^3 \|u\|_{H^2(\Omega)}^2$$

  Finaly we take the square root and get:

$$|||u - \pi u||| = |||e_I||| \leq Ch^{3/2} \|u\|_{H^2(\Omega)}$$

3.4 Final result

Using the above results we can apply the triangle inequality to get:

$$|||u - u_h||| \leq |||u - \pi u||| + |||\pi u - u_h||| \leq Ch^{3/2} \|u\|_{H^2(\Omega)}.$$

This gives us the final error estimate:

Theorem 3.1 (Theorem 10.1 in Larson/Bengzon)**.**
The finite element solution u_h, defined by (4)*, satisfies the estimate*

$$|||u - u_h||| \leq Ch^{3/2} \|u\|_{H^2(\Omega)}$$

4 Computer Implementation

See Larson/Bengzon for the simulation code. The only thing to be aware of is that the book uses the outdated pdecont(p,t,U) function, which has to be replaced with the new pdeplot(p,e,t) function.

Heat transfer in a fluid flow

This implementation example considers a channel with a circular obstacle (a hot cylinder) and a prescribed incompressible velocity field b.

- Temperature u solves a transport equation of the form

$$-\varepsilon \Delta u + b \cdot \nabla u = 0 \quad \text{in } \Omega,$$

  with mixed boundary conditions:

  - inflow: $u = 0$ (cold fluid);
  - cylinder: $u = 1$ (hot object);
  - walls: no normal heat flux (insulated);
  - outflow: convective-type boundary condition.

Dirichlet conditions are approximated by large Robin coefficients (e.g. $\kappa \approx 10^6$). The reason for using Robin coefficients as apposed to Dirichlet and Neumann is only to simplify the code implementation using the old code methods from the book. The weak form is derived by integrating both diffusion and convection terms by parts and collecting boundary terms via suitable functions κ, g_D, g_N. The GLS term $\delta(b \cdot \nabla u_h, b \cdot \nabla v_h)$ is then added to the left-hand side.

Using the books code we get the following output:

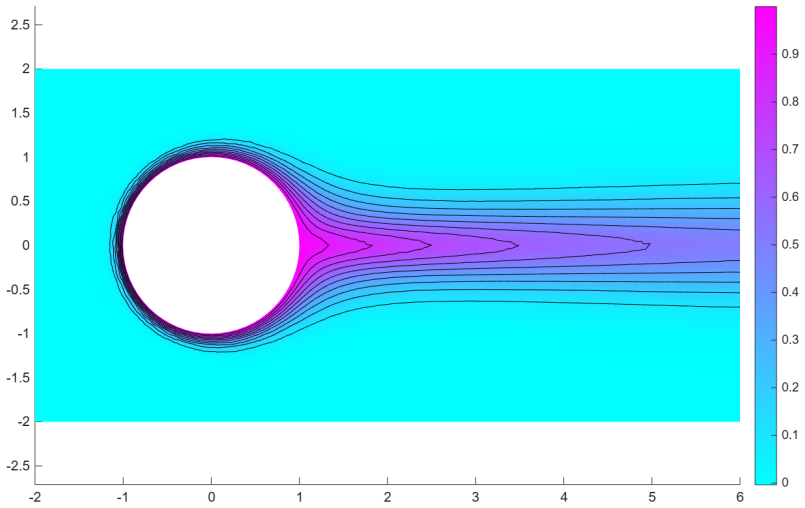


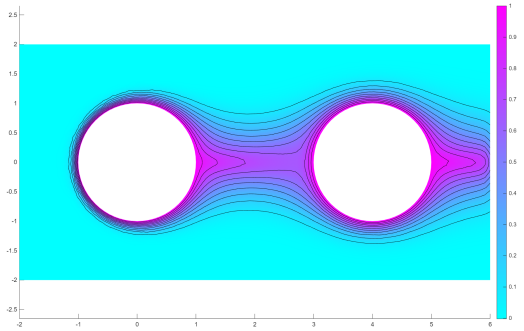
Figure 1: Larson/Bengzon book simulation using $\varepsilon = 0.01$ and $h = 0.1$

Observed behaviour of the numerical solution

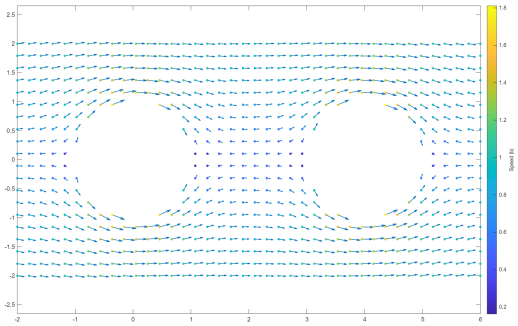
For a diffusion parameter ε significantly smaller than the mesh size h (here $\varepsilon = 0.01$ and $h = 0.1$), a standard Galerkin method would show strong oscillations. With the GLS method, we can see that boundary layers resolved in a stable way, smooth downstream transport of heat behind the cylinder, no non-physical overshoots or oscillations and the isocontours of u_h exit the domain without strong distortion.

Adding multiple objects

We can changing the code in the geometry function to edit the geometry of the simulation. One thing that could be interesting would be to see what happens if you where to add multiple hot object. On the following figure we see how the flow around 2 cylinders changes, ignoring that the water flow might be oversimplified here.



(a) 2 objects using $\varepsilon = 0.01$ and $h = 0.1$



(b) Associated glyphs showing the fluid velocity field b

Figure 2: Adding two cylinders to simulation

Here we can see that this changes the simulation downstream of the first cylinder. We see that the value is lowest near the inlet/left side. It looks a lot like the situation with one cylinder, but where the temperature further increases downstream after the second cylinder.

Here is the generalized code that can be used to add an arbitrary number of object to the simulation. Note however that the functions gD2, FlowField and Kappa2 have to be changed accordingly for the boundary conditions to be applied correctly:

```
function g = RectCircg2()
    % Rectangle: [-2,6] x [-2,2]
    % 3 = rectangle, 4 = number of edges,
    R1 = [3; 4; ...
          -2; 6; 6; -2; ...
          -2; -2; 2;  2];

    % First circle: center (0,0), radius 1
    C1 = [1; 0; 0; 1];

    % Second circle: center (4,0), radius 1
    C2 = [1; 4; 0; 1];

    % Pad circles to same length as rectangle
    C1 = [C1; zeros(length(R1) - length(C1), 1)];
    C2 = [C2; zeros(length(R1) - length(C2), 1)];

    % Geometry description matrix
    gd = [R1, C1, C2];

    % Names for the objects
    ns = char('R1', 'C1', 'C2')';

    % Set formula: rectangle minus both circles
    sf = 'R1-C1-C2';

    % Decomposed geometry matrix returned to initmesh
```

```
    g = decsg(gd, sf, ns);
end
```

5 References

Main book:

- Larson, M. G. and Bengzon, F. The Finite Element Method: Theory, Implementation, and Applications. Springer, 2013.

Further Reading:

- C. Johnson. Numerical Solution of Partial Differential Equations by the Finite ElementMethod. Studentlitteratur, 1987
- H. G. Roos, M. Stynes, and L. Tobiska. Numerical Methods for Singularly Perturbed Differential Equations. Springer, 1996